M.I.N.E.R.S. Rover - Dual BTS7960 Motor Drivers Wiring Guide

Configuration: 2x BTS7960 (IBT-2) 43A High-Power Drivers (Left Side Driver & Right Side Driver), Dedicated 12V Battery, 4WD DC Motors, Arduino Mega 2560

CRITICAL SAFETY RULE: COMMON GROUND

The 12V Battery Negative Terminal (B-) MUST be connected to BOTH Motor Drivers' B- slots AND to an Arduino Mega GND pin for a shared common ground reference.

1. Power Terminals (12V Battery $ ightarrow$ Both Motor Drivers)

Driver Module	Terminal Slot	Connection Target	Cable Type & Details
DRIVER 1 (Left)	B+	12V Battery POSITIVE (+)	Thick Power Cable (16 AWG)
DRIVER 1 (Left)	B-	12V Battery NEGATIVE (-)	Thick Power Cable (Connect to Arduino GND!)
DRIVER 2 (Right)	B+	12V Battery POSITIVE (+)	Thick Power Cable (16 AWG)
DRIVER 2 (Right)	B-	12V Battery NEGATIVE (-)	Thick Power Cable (Connect to Arduino GND!)

2. Motor Outputs (Connecting 4 Motors across 2 Drivers)

Driver Module	Output Slots	Connected Motors	Wiring Setup
DRIVER 1 (Left Side)	M+ & M-	Front-Left Motor & Rear-Left Motor	Wire Both Left Motors in Parallel to Driver 1
DRIVER 2 (Right Side)	M+ & M-	Front-Right Motor & Rear-Right Motor	Wire Both Right Motors in Parallel to Driver 2

3. Control Signal Pins (Arduino Mega 2560 $ ightarrow$ Both Motor Drivers)

BTS7960 Header Pin	DRIVER 1 (Left Side)	DRIVER 2 (Right Side)	Function / Purpose
VCC	Arduino Mega 5V	Arduino Mega 5V	Logic 5V Power for Optocoupler Isolator
GND	Arduino Mega GND	Arduino Mega GND	Logic Signal Ground Reference
R_EN (Right Enable)	Arduino Pin D30 (or 5V)	Arduino Pin D32 (or 5V)	Forward Half-Bridge Enable
L_EN (Left Enable)	Arduino Pin D30 (or 5V)	Arduino Pin D32 (or 5V)	Reverse Half-Bridge Enable
RPWM (Right PWM)	Arduino Pin D2 (PWM)	Arduino Pin D4 (PWM)	Forward Speed PWM Signal (0-255)
LPWM (Left PWM)	Arduino Pin D3 (PWM)	Arduino Pin D5 (PWM)	Reverse Speed PWM Signal (0-255)
R_IS & L_IS	LEAVE UNCONNECTED	LEAVE UNCONNECTED	Current Sensing Output (Not Needed)

Motor Direction Check

If the Left or Right wheels rotate backward when driving FORWARD ('CMD:F' or 'W'), simply swap M+ and M- wires for that specific driver module!